

EduGenie – Google Gemini Powered Learning Assistant

1. Project Overview

EduGenie is a Google Gemini powered learning assistant designed to help students learn and understand different topics more easily. The application provides AI-powered educational support through multiple learning features.

2. Problem Statement

Students may find it difficult to understand complex topics, obtain quick explanations, revise large amounts of information, and decide what to learn next. EduGenie provides these learning activities through a single application.

3. Objectives

The main objectives of EduGenie are:

- To provide answers to students' questions.
- To explain complex concepts in simple language.
- To generate quizzes for practice.
- To summarize educational text.
- To recommend suitable learning paths.
- To provide an easy-to-use learning assistant.

4. Main Features

Question and Answer

Students can enter questions and receive AI-generated answers.

Simplified Explanation

Complex concepts can be explained in a simpler and more understandable way.

Quiz Generation

EduGenie can generate quiz questions based on a selected topic.

Text Summarization

The application can summarize longer text and provide the important information in a concise form.

Learning Path Recommendation

EduGenie can provide learning recommendations based on a topic.

5. Technologies Used

- Python
- FastAPI
- Uvicorn
- HTML
- CSS
- Jinja2
- Google Gemini API
- Visual Studio Code
- GitHub

6. System Architecture

EduGenie uses a frontend and backend architecture. The user interacts with the web interface. The FastAPI backend receives the request and sends the appropriate task to the corresponding learning module. The required AI processing is performed using the Google Gemini API, and the result is returned to the user.

7. Project Modules

The main Python modules are:

- `main.py` – main application and routing.
- `qna.py` – Question and Answer functionality.
- `explanation_module.py` – simplified explanation functionality.
- `quiz_module.py` – quiz generation.
- `summary_module.py` – text summarization.
- `learning_path.py` – learning path recommendations.

8. Application Workflow

1. The user opens EduGenie.
2. The user selects a learning task.
3. The user provides the required input.
4. The request is sent to the FastAPI backend.
5. The appropriate module processes the request.
6. Google Gemini provides AI-generated content where required.
7. The result is displayed to the user.

9. Testing

The major EduGenie features are tested using different sample inputs. Testing covers Question and Answer, Simplified Explanation, Quiz Generation, Text Summarization, and Learning Path Recommendation.

10. Future Enhancements

Possible future enhancements include:

- Voice-based interaction.
- Multilingual learning support.
- Mobile application support.
- Student progress tracking.
- Gamification.
- Image and PDF based learning.
- Integration with learning management systems.

11. Conclusion

EduGenie provides an AI-powered platform that combines several useful learning functions in one application. It is designed to support students with explanations, answers, quizzes, summaries and learning recommendations.